

Real-World Data Linear Regression

Least-squares fit of global temperature anomaly on atmospheric CO₂

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Section 3H

Observations 15

Source NASA GISS + Mauna Loa

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Results at a glance

Least-squares line fitted by hand formulas (no polyfit).

0.011537 °C/ppm Slope a₁	-3.761204 °C Intercept a₀	0.066066 °C² SSE
0.973542 r²	0.071288 °C Std error	

$\hat{y} = -3.761204 + 0.011537 \cdot x$ (y = temperature anomaly °C, x = CO₂ in ppm)

Data and source

Source: NASA Goddard Institute for Space Studies (GISS) surface-temperature anomalies and Mauna Loa (NOAA) CO₂ record.

NO.	X · CO ₂ (PPM)	Y · TEMP ANOMALY (°C)
1	316.9	0.01
2	320.0	-0.08
3	325.7	0.02
4	331.1	-0.06
5	338.7	0.21
6	346.0	0.11
7	354.4	0.38
8	360.9	0.45
9	369.5	0.42
10	379.7	0.62
11	389.9	0.72
12	400.8	0.86
13	408.5	0.97

NO.	X · CO ₂ (PPM)	Y · TEMP ANOMALY (°C)
14	414.7	0.98
15	424.6	1.21

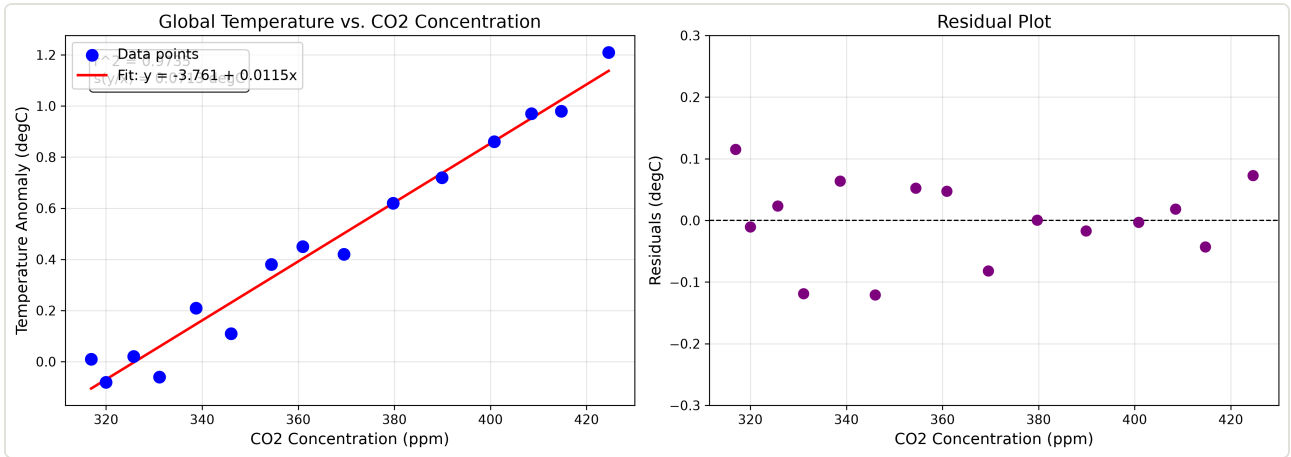
Least-squares computation

Hand formulas — no numpy.polyfit.

QUANTITY	FORMULA	VALUE
n	count of points	15
Σx	Σx	5481.4000
Σy	Σy	6.8200
Σxy	Σ(x·y)	2702.9180
Σx ²	Σ(x ²)	2021313.6600
a ₁	(nΣxy - ΣxΣy) / (nΣx ² - (Σx) ²)	0.011537 °C/ppm
a ₀	(Σy - a ₁ Σx) / n	-3.761204 °C
SSE	Σ(y - ŷ) ²	0.066066 °C ²
SS _t	Σ(y - ȳ) ²	2.496973 °C ²
r ²	1 - SSE/SS _t	0.973542
s _{y/x}	√(SSE / (n - 2))	0.071288 °C

Graphs

Generated by running Cabudsan_DaphneAnne_3H_Lab03.py — saved automatically as regression_results.png.



How the graph is produced

The script builds a one-row, two-column figure. Left: scatter of data plus the fitted line with an r^2 / $s_{y/x}$ text box. Right: residuals against x with a zero line.

`plt.savefig('regression_results.png', dpi=300, bbox_inches='tight')` writes the PNG in the same folder as the script.

Prediction

Predict temperature anomaly for CO₂ = 450 ppm using the fitted line.

$$\hat{y} = -3.761204 + 0.011537 \times 450 = -3.761204 + 5.191650 = 1.430 \text{ }^{\circ}\text{C}$$

IN PLAIN WORDS

A CO₂ concentration of 450 ppm is predicted to give a temperature anomaly of **1.430 °C** above the 1951–1980 baseline. Because 450 ppm is near the top of the observed range, this is a mild extrapolation and is reasonably reliable.

Interpretation

$r^2 = 0.9735$ means about 97.4% of the variation in temperature anomaly is explained by CO₂.

- **Slope ($a_1 = 0.011537 \text{ }^{\circ}\text{C/ppm}$):** each 1 ppm rise in CO₂ is associated with about 0.0115 °C of warming — roughly 1.15 °C per 100 ppm.
- **Intercept ($a_0 = -3.761204 \text{ }^{\circ}\text{C}$):** value at CO₂ = 0 ppm. **Not physically meaningful** — it extrapolates far outside the data (316.9–424.6 ppm), and CO₂ has never been near zero in recent Earth history.
- **$r^2 = 0.973542$ (very good fit):** the linear model explains ~97.4% of the variability; only ~2.6% is due to other factors. $r^2 = 1$ would be a perfect fit.
- **Standard error $s_{y/x} = 0.071288 \text{ }^{\circ}\text{C}$:** typical miss of the line from the data — about $\pm 0.07 \text{ }^{\circ}\text{C}$.
- **Residual plot:** residuals scatter roughly randomly around zero with no obvious U-shape or fan-out, supporting the linear model and roughly constant variance.

Python code

Full script — Cabudsan_DaphneAnne_3H_Lab03.py

```
import numpy as np
import matplotlib.pyplot as plt

# 1. DATA ENTRY
x = np.array([316.9, 320.0, 325.7, 331.1, 338.7, 346.0, 354.4,
              360.9, 369.5, 379.7, 389.9, 400.8, 408.5, 414.7, 424.6])
y = np.array([0.01, -0.08, 0.02, -0.06, 0.21, 0.11, 0.38,
              0.45, 0.42, 0.62, 0.72, 0.86, 0.97, 0.98, 1.21])
n = len(x)

# 2. LEAST SQUARES BY FORMULA (no polyfit)
sum_x = np.sum(x)
sum_y = np.sum(y)
sum_xy = np.sum(x * y)
sum_x2 = np.sum(x ** 2)

a1 = (n*sum_xy - sum_x*sum_y) / (n*sum_x2 - sum_x**2) # slope
```

```
a0 = (sum_y - a1*sum_x) / n                                # intercept

# 3. GOODNESS OF FIT
y_pred  = a0 + a1 * x
residuals = y - y_pred
SSE = np.sum(residuals ** 2)
y_mean = np.mean(y)
SSt = np.sum((y - y_mean) ** 2)
r_squared = 1 - (SSE / SSt)
syx = np.sqrt(SSE / (n - 2))

print(f"a1 = {a1:.6f}, a0 = {a0:.6f}")
print(f"SSE = {SSE:.6f}, SSt = {SSt:.6f}")
print(f"r2 = {r_squared:.6f}, standard error = {syx:.6f}")

# 4. PREDICTION
x_pred = 450.0
print(f"Prediction at {x_pred} ppm: {a0 + a1*x_pred:.3f} degC")

# 5. VISUALIZATION
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))
ax1.scatter(x, y, color='blue', s=80, label='Data', zorder=5)
ax1.plot(x, y_pred, color='red', lw=2, label=f'Fit: y={a0:.3f}+{a1:.4f}x')
ax1.set_xlabel('CO2 (ppm)'); ax1.set_ylabel('Temp anomaly (degC)')
ax1.set_title('Temp vs CO2'); ax1.grid(alpha=0.3); ax1.legend()
ax2.scatter(x, residuals, color='purple', s=60)
ax2.axhline(0, color='black', ls='--')
ax2.set_xlabel('CO2 (ppm)'); ax2.set_ylabel('Residuals (degC)')
ax2.set_title('Residual plot'); ax2.grid(alpha=0.3); ax2.set_ylim(-0.3, 0.3)
plt.tight_layout()
plt.savefig('regression_results.png', dpi=300, bbox_inches='tight')
plt.show()
```

Files to submit

FILE	PURPOSE
Cabudsan_DaphneAnne_3H_Lab03.py	Python script (this report is generated from it)
regression_results.png	Graphs produced by the script
Cabudsan_DaphneAnne_3H_Lab03.pdf	Final report (Surname_Firstname_3H_Lab03.pdf)